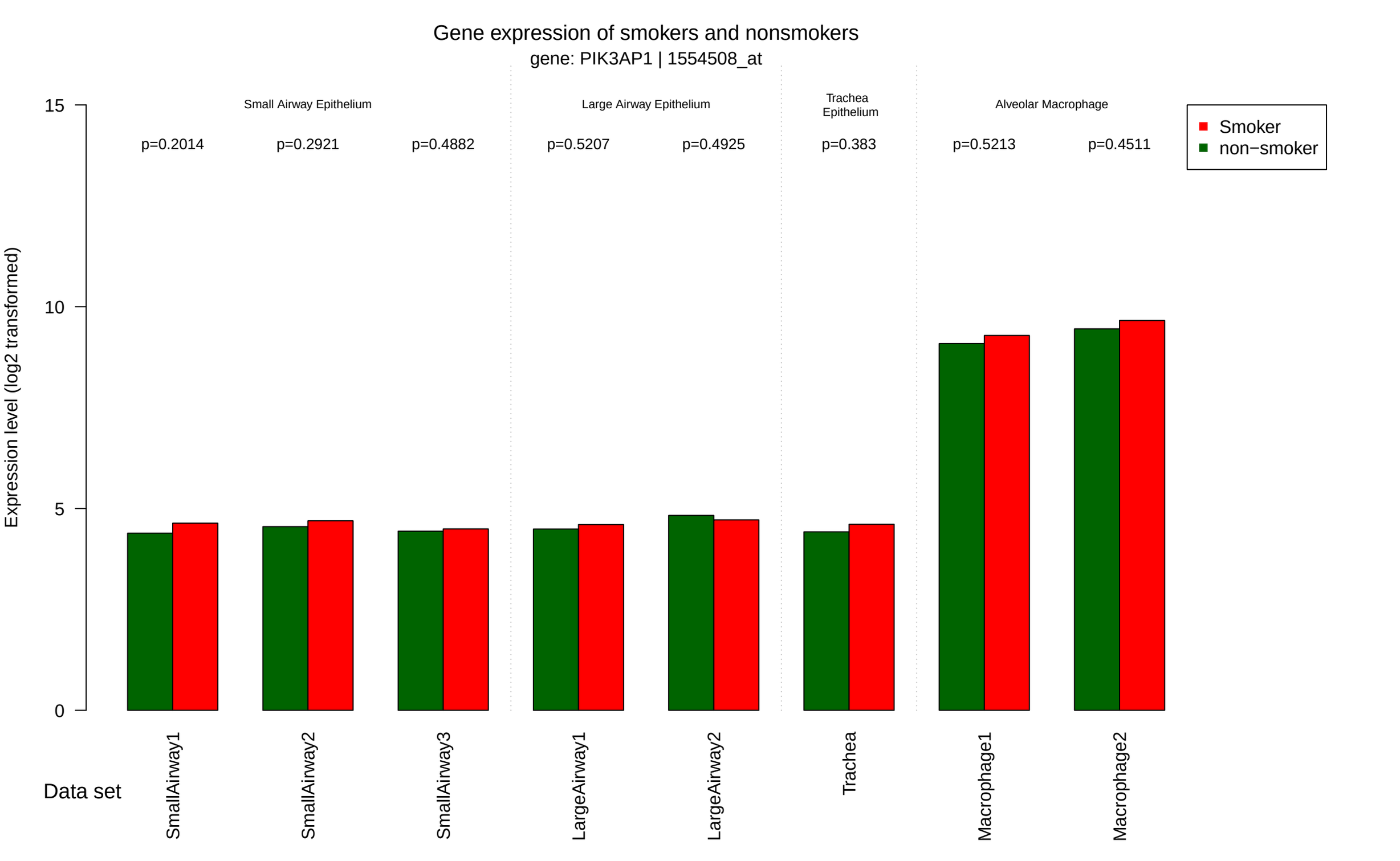
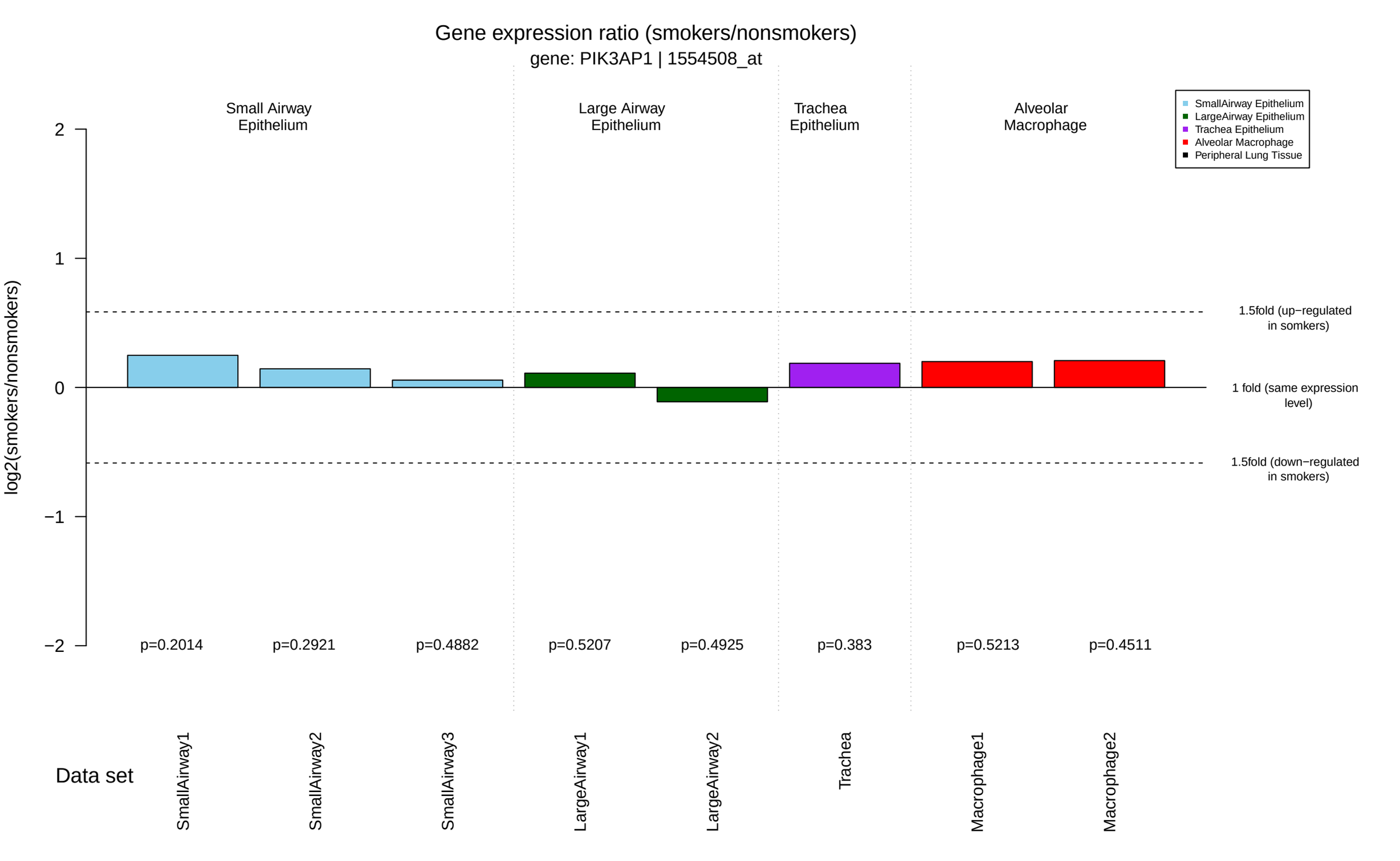
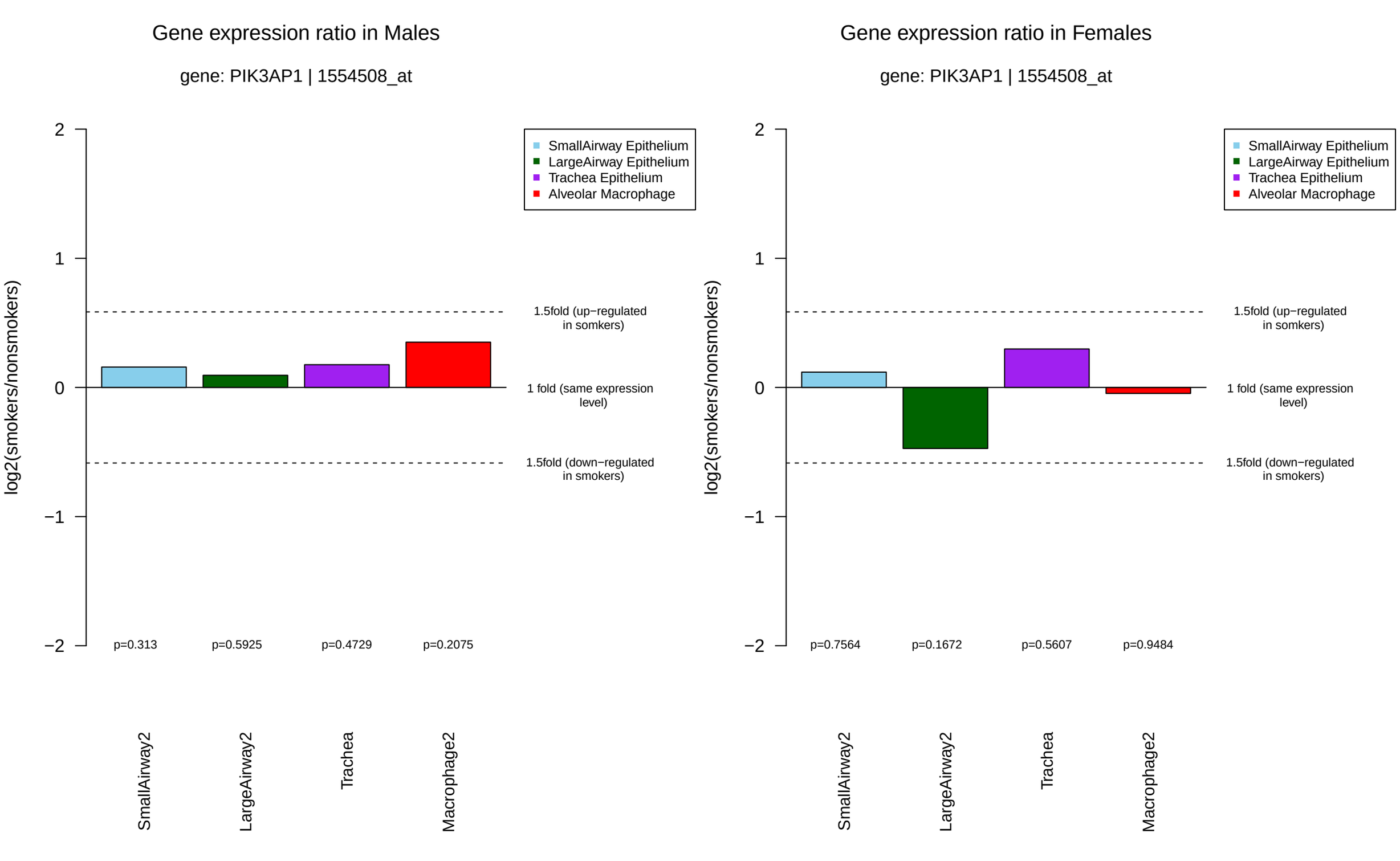
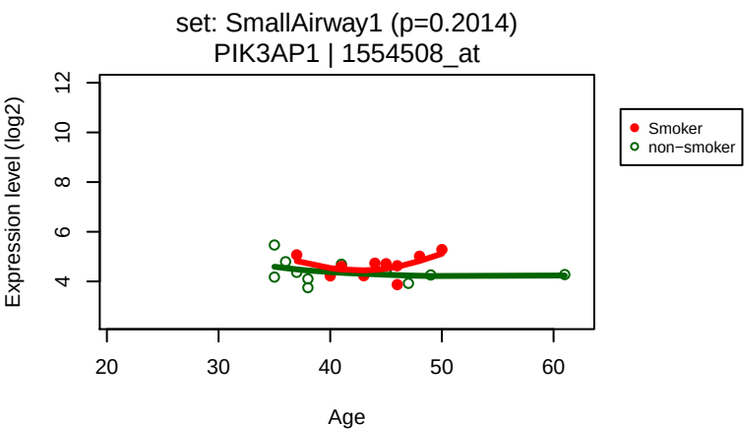
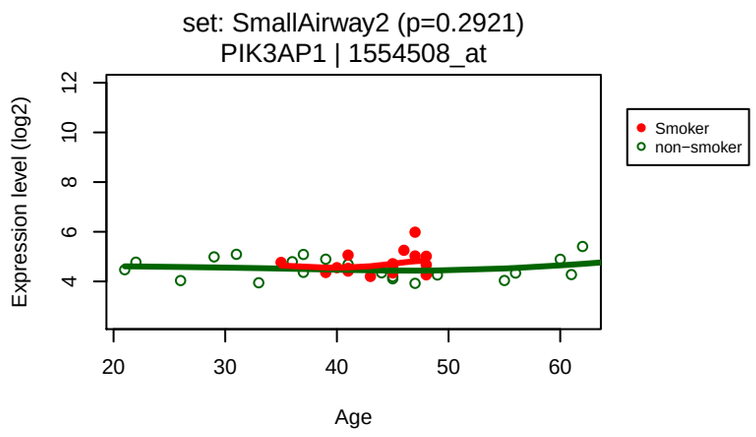
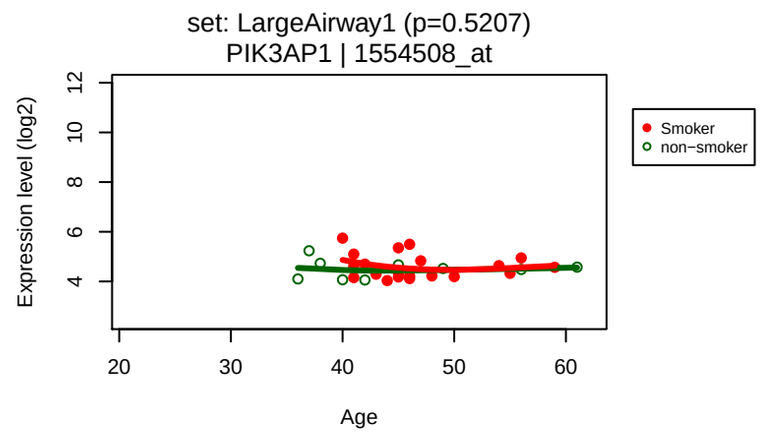

Gene expression of smokers and nonsmokers
gene: PIK3AP1 | 1554508_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2014
p=0.2921
p=0.4882
p=0.5207
p=0.4925
p=0.383
p=0.5213
p=0.4511
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

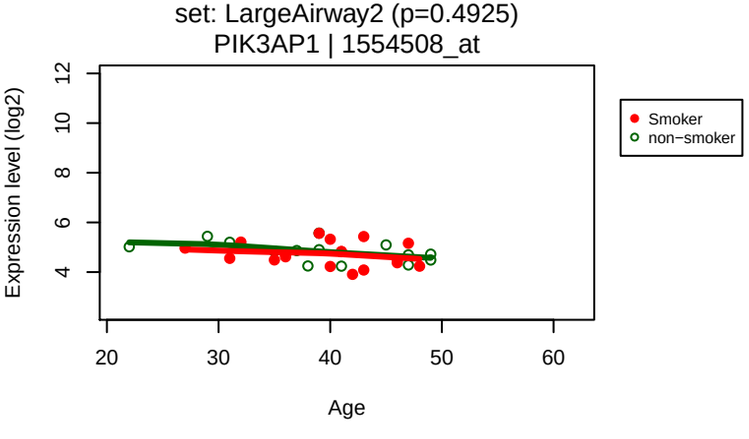

Gene expression ratio (smokers/nonsmokers)
gene: PIK3AP1 | 1554508_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2014
p=0.2921
p=0.4882
p=0.5207
p=0.4925
p=0.383
p=0.5213
p=0.4511
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

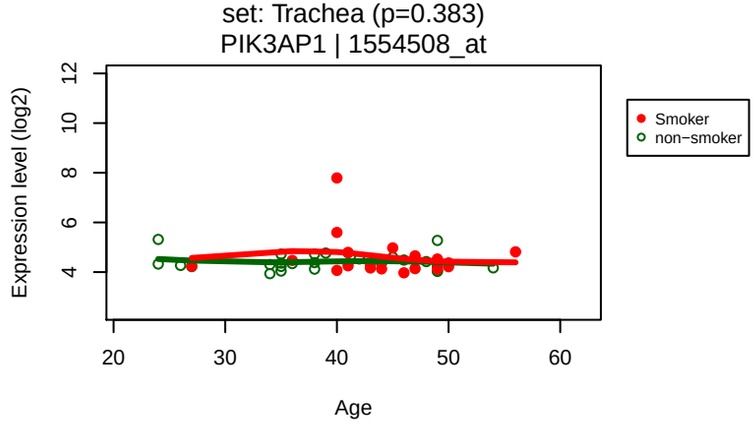

Gene expression ratio in Males
gene: PIK3AP1 | 1554508_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.313
p=0.5925
p=0.4729
p=0.2075
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PIK3AP1 | 1554508_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7564
p=0.1672
p=0.5607
p=0.9484
SmallAirway2
LargeAirway2
Trachea
Macrophage2

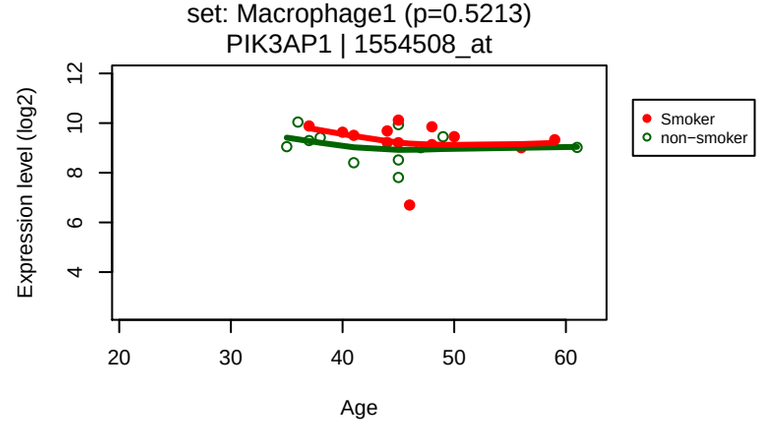

set: SmallAirway1 (p=0.2014)
PIK3AP1 | 1554508_at
Expression level (log2)
Age
Smoker
non−smoker

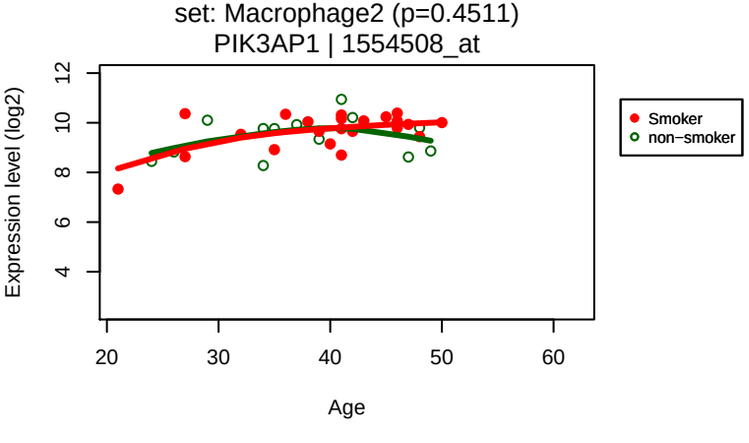

set: SmallAirway2 (p=0.2921)
PIK3AP1 | 1554508_at
Expression level (log2)
Age
Smoker
non−smoker

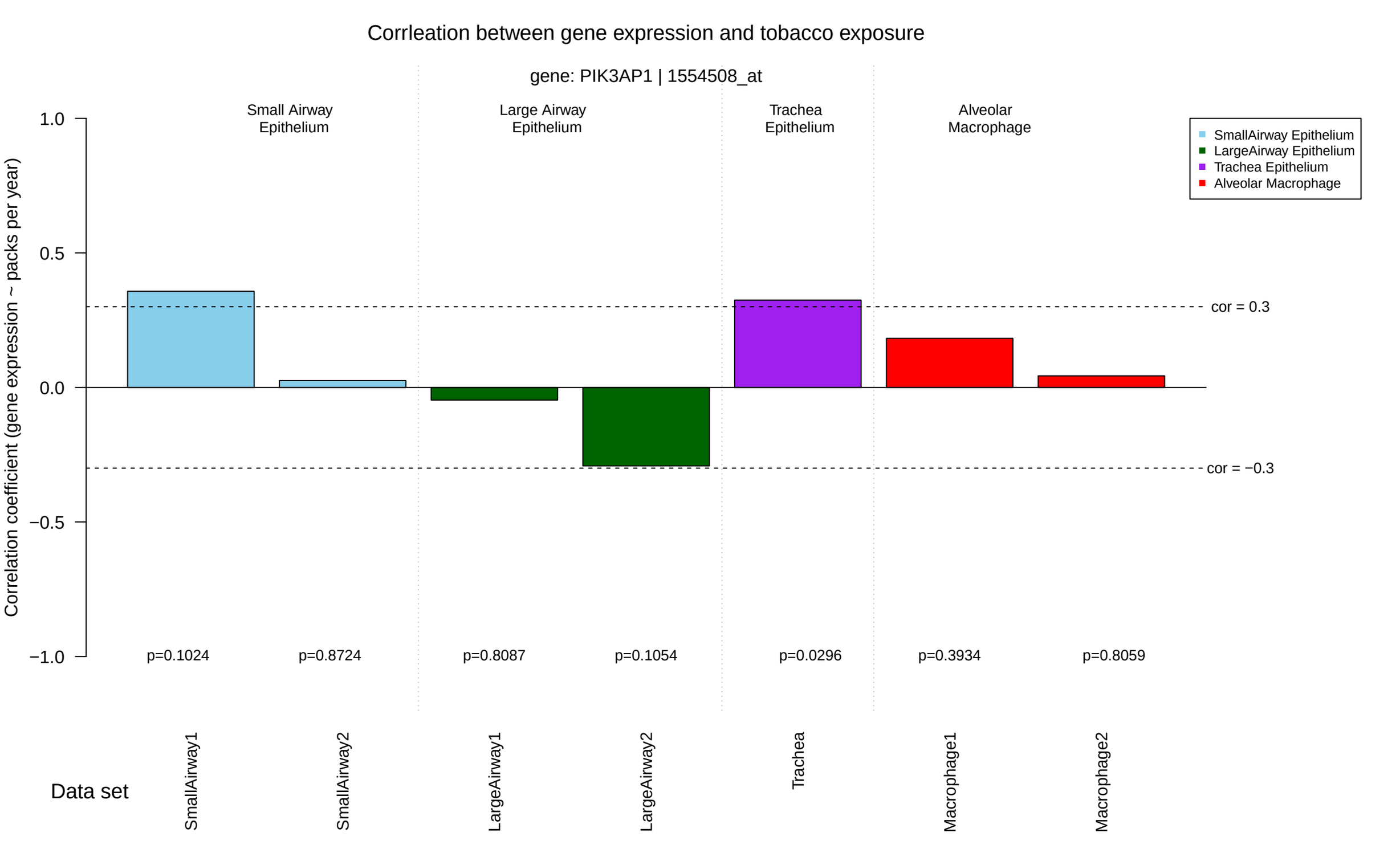

set: LargeAirway1 (p=0.5207)
PIK3AP1 | 1554508_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4925)
PIK3AP1 | 1554508_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.383)
PIK3AP1 | 1554508_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5213)
PIK3AP1 | 1554508_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4511)
PIK3AP1 | 1554508_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PIK3AP1 | 1554508_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1024
p=0.8724
p=0.8087
p=0.1054
p=0.0296
p=0.3934
p=0.8059
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2